```
pswayam@pswayam-VirtualBox:~$ apt-get install php-pear
```

As a next step, we need to create a set of directories required to successfully run Snort. One can use the following commands to create these files:

```
pswayam@pswayam-VirtualBox:~$ mkdir /etc/snort
pswayam@pswayam-VirtualBox:~$ mkdir /etc/snort/rules
pswayam@pswayam-VirtualBox:~$ mkdir /var/log/snort
```

The *snort.conf* file controls everything about what Snort watches, how it defends itself from attack, what rules it uses to find malicious traffic, and even how it watches for potentially dangerous traffic that isn't defined by a signature. A very good understanding of what is in this file and how it can be configured is pretty essential for successful deployment of Snort as an IDS. By default, this configuration file is located @ */etc/snort*. If the configuration is located somewhere else, then you need to specify a *–c* switch along with the file's location.

Alerts are placed in the *Alert* file in your logging directory (*/var/log/snort* by default, or the directory you specify with the *-l* switch). Snort exits with an error if the configuration file or its logging directory does not exist. You can specify what type of alert you want (full, fast, none, or UNIX sockets) by supplying the *-A* command-line switch. You need to use the *-A* switch if you do not use the *-c* switch, or Snort will not enter alert mode. Additionally, if you use just the *-c* switch, Snort will use the full alert mode by default. The *snort.conf* file is used to store all the configuration information for a Snort process running in IDS mode. The majority of the *snort.conf* file's content is commented-out instructions on how to use the file. Some of the popular configurable items of this *snort.conf* file are listed in Table 1.

Table 1

	Variable	Description
1	HOME_NET	Specifies the IP address of the system that you are protecting. So when you use Snort, you will need to change this parameter to your actual home network IP address range and do not leave this to the default value of 'any'.
2	ORACLE_PORTS	Used to specify the port that Oracle is listening on.
3	RULE_PATH	Points to the location of the rule sets in your file system. Typically, rules are stored in */etc/snort/rules*. So please make sure to use the full path name or whatever the right location is on your system.

Please note here that to get Snort ready to run, one needs to change the default configuration settings to match your local environment and operational preferences. Also, if you want to change the settings of the default configuration file, you need to have root privileges.

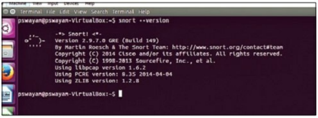

Figure 1: Checking if the Snort installation was successful

It is very common to see *include* items in Snort configuration files. These are commands that instruct Snort to include the information in files located in Snort's sensor's file system. Typically, these files contain configuration related information and also the files that contain rules which Snort can use to catch bad behaviour. As expected, the actual rules themselves are not entered directly into the configuration file to simplify management.

One can set an alert in the rules files with the following structure:

```
<Rule Actions> <Protocol> <Source IP Address> <Source Port>
<Direction Operator> <Destination IP Address> <Destination port
> (rule options)
```

Let us take a quick look at the rule structure with an example of each:

Structure	Example
Rule action	alert
Protocol	ICMP
Source IP address	any
Source port	any
Direction operator	->
Destination IP address	any
Destination port	any
Rule options	(msg:"ICMP Packet"; sid:477; rev:3;)

It is important to understand here that a few lines are added for each alert and these lines carry the following information:

a. IP address of the source
b. IP address of the destination
c. Packet type and useful header information

In order to execute Snort, one can use the following command:

```
pswayam@pswayam-VirtualBox:~$ snort -c /etc/snort/snort.conf -l
/var/log/snort/
```

Now, a file called *Alert* will get created in the */var/log/snort*